

Warehouse Management System - Project Report

1. Introduction

The Warehouse Management System (WMS) is designed to efficiently track and manage inventory across multiple warehouses. This system helps businesses streamline operations by centralizing data related to suppliers, products, inventory levels, and storage facilities.

2. Abstract

This project involved designing a relational database schema, populating it with sample data, and implementing key SQL features such as queries, triggers, and stored procedures. The system provides real-time insights into stock levels, generates reorder alerts, and allows seamless stock transfers.

3. Tools Used

- Database: MySQL
- Development Environment: DBeaver (SQL GUI)
- Programming Support: Faker (Python) for data generation

4. Steps Involved in Building the Project

a. Schema Design:

- Tables: suppliers, warehouses, products, inventory
- Relationships defined via foreign keys

b. Sample Data Insertion:

- 10 suppliers, 5 warehouses, 30 products, and 100 inventory records inserted using synthetic data

c. SQL Queries:

- Stock level checks across warehouses
- Reorder alerts for low-stock products

d. Trigger:

- `trg_low_stock_alert`: Automatically inserts a record into `low_stock_alerts` when stock falls below reorder level

e. Stored Procedure:

- `transfer_stock`: Transfers quantity of a product from one warehouse to another, ensuring stock validation

5. Conclusion

The project successfully demonstrates a functional backend for warehouse inventory tracking using SQL. With scalable design and automation via triggers and procedures, the system enhances operational efficiency and ensures data integrity. This backend can serve as the foundation for future integration with frontend dashboards or inventory apps.